

Merging Blockchain and Augmented Reality for an Immersive Traceability Platform

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ABSTRACT

The demand for ethically sourced and safe products has surged, prompting industries to adopt intricate traceability systems. Blockchain technology revolutionizes traceability by ensuring data integrity and transparency in supply chains. However, complexities within supply chains often obfuscate meaningful insights for consumers. This paper explores leveraging Augmented Reality to enhance Blockchain-based traceability systems. By integrating AR, consumers can seamlessly access traceability information through QR codes, presented via optimized 3D models. This immersive approach fosters trust by visually demonstrating product quality. The architecture combines QR codes, Vuforia markers, and Blockchain, ensuring data security and immutability.

Index Terms: Computer systems organization—Distributed architectures—;—Software and its engineering—Software design engineering—Security and privacy—Security services—

1 INTRODUCTION

Consumer expectations for ethically sourced, environmentally friendly, and safe products have increased the popularity of traceability systems. In response, industries have incorporated sophisticated tracking mechanisms to monitor each stage of production and distribution. Blockchain technology, with its decentralized and immutable ledger, is at the forefront of enhancing traceability systems. Blockchain ensures data integrity and transparency by recording every transaction and event along the supply chain. However, complex supply chains can lead to convoluted data structures, making it difficult for consumers to glean meaningful insights. On the other side, end-users in the future will have a continuously greater opportunity to interact with the objects in their natural environment in an immersive way through Augmented Reality applications. Novel headsets such as Apple Vision Pro and Meta Quest Pro promise to change the way people will accomplish their everyday tasks like shopping at the supermarket. This paper aims to explore Blockchain-based traceability systems, using AR to improve the accessibility of such platforms, giving the final consumer the possibility to scan a QR code and easily access only the needed traceability information in a visually optimized way. The product is shown to the users using a 3D model that modifies its shape by following a given quality, improving their trust in the producer.

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2 STATE OF THE ART

Blockchain is a decentralized, immutable ledger that securely and openly stores data and transactional information. Blockchain can create verifiable and tamper-proof digital assets thanks to its inner characteristics of data immutability, usage of consensus mechanisms and transparency. Augmented Reality (AR) is a transformative technology that overlays digital information, such as virtual objects or contextual data, onto the real world. QR codes and markers have redefined AR interactions, while Blockchain technology assures data integrity and transparency. Integrating these technologies in AR enhances information dissemination and user engagement. Current literature lacks on a deep analysis regarding AR and Blockchain technology in the traceability field. Some architectures based on decentralized platforms have been proposed in different topics, from face-to-face collaboration using a web based XR framework [4] to the creation of virtual environments using Metaverse [5]. Blockchain and AR applications can be combined in various domains: education and training, advertising, tourism, virtual experiences [1]. The main contribution of this paper regards the possibility to give useful information about a product in an immersive way, giving consumers the possibility to scan a QR code and understand its quality simply by using their smartphone.

3 ARCHITECTURE

The architecture of the platform is shown in Fig. 1. The consumer scans a QR code on the product label and the front-end distributed application sends a request to load the 3D model of the scanned product that is visualized in AR using Vuforia Engine. At the same time, a request is sent to the public Blockchain (through a smart contract) to access the traceability information regarding that product. The obtained data is shown to the consumer on top of the generated 3D model.

By using QR codes and Vuforia markers as entry points to Blockchain-hosted content, developers can ensure data security and immutability. Vuforia markers can initiate dynamic AR displays of Blockchain-verified information, while QR codes could lead users to transaction histories or supply chain details stored on the Blockchain. The combination of QR codes and Vuforia markers can happen in different ways. One approach is to insert a marker inside the QR code. Different tracking techniques can lead to this goal: Vuforia gives the opportunity to use a VuMark that can be used to ensure a high number of feature points. However, this approach can lead to issues if two products are scanned simultaneously, because of the impossibility to distinguish between different products using the same marker. An alternative way is to directly use the QR code as a marker: it has lots of unique points and it can distinguish between two products. However, this scenario will require one QR code for each product uploaded in the Vuforia engine, so it could heavily affect the performance of the platform. The second scenario is chosen, due to the possibility to use Vuforia Cloud Recognition service to manage a huge number of image targets. Vuforia also grants the possibility to directly scan a QR code inside its environment using

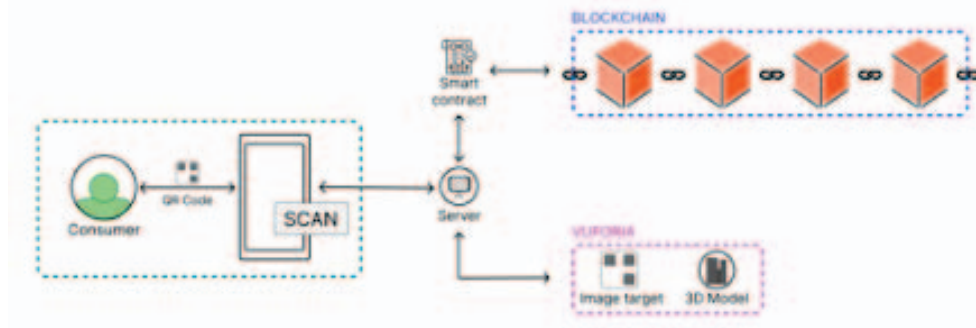


Figure 1: Architecture of the proposed platform

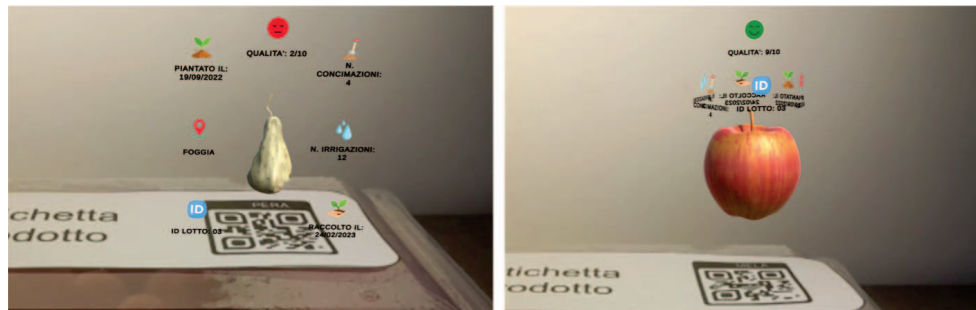


Figure 2: Two alternative graphical interfaces. (a) a pear, with quality 2/10, is surrounded by the attributes gathered from the Blockchain. (b) an apple, with quality 9/10, has its attributes rotating on top of it.

the Barcode Scanner service. This method is not compliant to the goal of the paper because it is limited to 2D objects, and it will not be possible to load 3D models on top of a scanned QR code.

4 SAMPLE SCENARIO

The platform has been tested in Unity. The sample scenario regards a traceability platform based on Blockchain technology, combining the benefits of both a public and a private Blockchain [3]. The platform must enable the general user to check the product origin, information, chemical-physical properties, and varietal origin [2]. Additionally, it must ensure that the product's agri-food chain is controlled and monitored by having access to data about the production phases, from seeding to selling, including all phases of any transformation. The project proposal specifically aims to experiment with the integration of Blockchain technology and, as a result, the real-time traceability of the information recorded on the history of the product.

First results show that the platform correctly reads the required information and shows them to the final user, together with the product quality shown on top of the gathered data. Additionally, the product changes its shape by following the given quality: the higher the quality, the better it will be seen by the final user. Fig. 2 shows two different ways to implement the graphical interface of the platform: one with fixed information around the product, all on the same plane; another one with animated information rotating on top of the product.

5 CONCLUSION

The increasing popularity of traceability systems in supply chains reflects the evolution of consumer expectations towards transparency and ethical consumption. Blockchain's integration in these systems brings immutable records and heightened trust. However, the chal-

lenge of enabling easy access to and comprehension of product information for consumers remains hard to accomplish. The possibility to use AR techniques helps in bridging this gap, giving consumers the possibility to easily read information thanks to immersive interfaces and platforms. Future works regard some additional analysis on the usage of QR codes as markers and testing scenarios when similar QR codes are used for different products and how the shape of the product label can infect the platform performance. A survey on consumers will also be done to evaluate the user experience in the context of Blockchain and AR used together in a traceability platform.

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